

## OUTSTANDING PROBLEMS

**(O1) The Andrews-Curtis conjecture.** Let  $F = F_n$  be the free group of a finite rank  $n$  with a fixed set  $X = \{x_1, \dots, x_n\}$  of free generators.

(a) A set  $Y = \{y_1, \dots, y_n\}$  of elements of  $F$  generates the group  $F$  as a normal subgroup if and only if  $Y$  is Andrews-Curtis equivalent to  $X$ , which means one can get from  $X$  to  $Y$  by a sequence of Nielsen transformations together with conjugations by elements of  $F$ .

(b) The same conjecture if one allows one more transformation, "stabilization", where  $X$  is extended to  $\{x_1, \dots, x_n, x_{n+1}, \dots\}$ , and its converse. [Background](#)

**(O2) The Burnside problem.** For what values of  $n$  are all groups of exponent  $n$  locally finite? Of particular interest are  $n=5$ ,  $n=8$ ,  $n=9$  and  $n=12$  - values for which, by the experts' opinion, groups of exponent  $n$  have a remote chance of being locally finite. [Background](#)

**(O3) Whitehead's asphericity problem.** Is every subcomplex of an aspherical 2-complex aspherical? Or, equivalently: if  $G = F/R = \langle x_1, \dots, x_n \mid r_1, \dots, r_m, \dots \rangle$  is an aspherical presentation of a group  $G$  (i.e., the corresponding relation module  $R/[R, R]$  is a free  $\mathbb{Z}G$ -module), is then every presentation of the form  $\langle x_1, \dots, x_n \mid r_{i_1}, \dots, r_{i_k}, \dots \rangle$ , aspherical as well? [Background](#)

**(O4)** The isomorphism problem for one-relator groups. [Background](#)

**(O5)** The conjugacy problem for one-relator groups. [Background](#)

**(O6)** Is every one-relator group without Baumslag-Solitar subgroups hyperbolic? [Background](#)

**(O7)** Let  $G$  be the direct product of two copies of the free group  $F_n$ ,  $n \geq 2$ , generated by  $\{x_1, \dots, x_n\}$  and  $\{y_1, \dots, y_n\}$ , respectively.

(a) Is it true that every generating system of cardinality  $2n$  of the group  $G$  is Nielsen equivalent to  $\{(x_1, 1), \dots, (x_n, 1), (1, y_1), \dots, (1, y_n)\}$ ?

(b) "Stabilized" version of the same question: Is it true that for every generating system of cardinality  $2n$ , there is  $k \geq 0$  such that this generating system, augmented by  $(x_{n+1}, 1), \dots, (x_{n+k}, 1), (1, y_{n+1}), \dots, (1, y_{n+k})$  is Nielsen equivalent to  $\{(x_1, 1), \dots, (x_{n+k}, 1), (1, y_1), \dots, (1, y_{n+k})\}$ ? [Background](#)

**\*(O8) Tarski's problems.** Let  $F = F_n$  be the free group of rank  $n$ ,  $\text{Th}(F)$  the elementary theory of  $F$ , i.e., all sentences of the group theory language which are true in  $F$ .

(a) Is it true that  $\text{Th}(F_2) = \text{Th}(F_3)$ ?

(b) Is  $\text{Th}(F)$  decidable? [Background](#)

**\*(O9) The Hanna Neumann conjecture.** If  $H$  and  $K$  are nontrivial subgroups of a free group, then  $\text{rank}(H \cap K) - 1 \leq (\text{rank}(H) - 1)(\text{rank}(K) - 1)$ . [Background](#)

**\*(O10)** Is the automorphism group of a free group of rank 2 linear? Or, equivalently, is the braid group  $B_4$  linear? [Background](#)

**(O11)** Is there an infinite finitely presented periodic group?

**(O12)** (a) **(I.Kaplansky)** Can the (integral) group ring of a torsion-free group have zero divisors? [Background](#)

(b) Is it true that the group of units of a group algebra  $kG$ , where  $G$  is torsion-free and  $k$  is a field, is generated by  $k^*$  and  $G$ ?